```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("ANALISIS DATA MINING BIKE BUYERS")
print("Berhasil memulai Google Colab")
```

```
ANALISIS DATA MINING BIKE BUYERS
Berhasil memulai Google Colab
```

```
from google.colab import files

uploaded = files.upload()
```

Pilih File bike_buyers.csv
bike_buyers.csv(text/csv) - 87182 bytes, last modified: 4/11/2026 - 100% done
Saving bike_buyers.csv to bike_buyers.csv

```
df = pd.read_csv('bike_buyers.csv')

print("TAMPILAN 10 DATA PERTAMA")
display(df.head(10))

print("\n--- INFORMASI STRUKTUR DATASET ---")
df.info()

print("\nJumlah Baris dan Kolom:")
print(df.shape)
```

TAMPILAN 10 DATA PERTAMA

	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
0	12496	Married	Female	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	42.0	No
1	24107	Married	Male	30000.0	3.0	Partial College	Clerical	Yes	1.0	0-1 Miles	Europe	43.0	No
2	14177	Married	Male	80000.0	5.0	Partial College	Professional	No	2.0	2-5 Miles	Europe	60.0	No
3	24381	Single	NaN	70000.0	0.0	Bachelors	Professional	Yes	1.0	5-10 Miles	Pacific	41.0	Yes
4	25597	Single	Male	30000.0	0.0	Bachelors	Clerical	No	0.0	0-1 Miles	Europe	36.0	Yes
5	13507	Married	Female	10000.0	2.0	Partial College	Manual	Yes	0.0	1-2 Miles	Europe	50.0	No
6	27974	Single	Male	160000.0	2.0	High School	Management	NaN	4.0	0-1 Miles	Pacific	33.0	Yes
7	19364	Married	Male	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	43.0	Yes
8	22155	NaN	Male	20000.0	2.0	Partial High School	Clerical	Yes	2.0	5-10 Miles	Pacific	58.0	No
9	19280	Married	Male	NaN	2.0	Partial College	Manual	Yes	1.0	0-1 Miles	Europe	NaN	Yes

```
--- INFORMASI STRUKTUR DATASET ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                1000 non-null   int64
 1   Marital Status    993 non-null    object
 2   Gender            989 non-null    object
 3   Income            994 non-null    float64
 4   Children          992 non-null    float64
 5   Education         1000 non-null   object
 6   Occupation        1000 non-null   object
 7   Home Owner        996 non-null    object
 8   Cars              991 non-null    float64
 9   Commute Distance  1000 non-null   object
 10  Region            1000 non-null   object
 11  Age               992 non-null    float64
 12  Purchased Bike    1000 non-null   object
```

```
print("--- JUMLAH DATA KOSONG SEBELUM ---")
```

```
print(df.isnull().sum())

# Mengisi data kosong pada kolom numerik dengan median
kolom_numerik = ['Income', 'Children', 'Cars', 'Age']

for kolom in kolom_numerik:
    df[kolom] = df[kolom].fillna(df[kolom].median())

# Mengisi data kosong pada kolom kategori dengan modus
kolom_kategori = ['Marital Status', 'Gender', 'Home Owner']

for kolom in kolom_kategori:
    df[kolom] = df[kolom].fillna(df[kolom].mode()[0])

print("\n--- JUMLAH DATA KOSONG SETELAH ---")
print(df.isnull().sum())
```

```
--- JUMLAH DATA KOSONG SEBELUM ---
ID                   0
Marital Status       7
Gender              11
Income               6
Children             8
Education            0
Occupation           0
Home Owner           4
Cars                 9
Commute Distance     0
Region               0
Age                  8
Purchased Bike       0
dtype: int64

--- JUMLAH DATA KOSONG SETELAH ---
ID                  0
Marital Status      0
Gender              0
Income              0
Children            0
Education           0
Occupation          0
Home Owner          0
Cars                0
Commute Distance    0
Region              0
Age                 0
Purchased Bike      0
dtype: int64
```

```
# Menghapus kolom ID
df.drop('ID', axis=1, inplace=True)

# Menampilkan data setelah kolom ID dihapus
print("Tampilan data setelah kolom ID dihapus:")
display(df.head())

# Menampilkan jumlah kolom
print("\nJumlah baris dan kolom:")
print(df.shape)
```

```
Tampilan data setelah kolom ID dihapus:
```

	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance	Region	Age	Purchased Bike
0	Married	Female	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	42.0	No
1	Married	Male	30000.0	3.0	Partial College	Clerical	Yes	1.0	0-1 Miles	Europe	43.0	No
2	Married	Male	80000.0	5.0	Partial College	Professional	No	2.0	2-5 Miles	Europe	60.0	No
3	Single	Male	70000.0	0.0	Bachelors	Professional	Yes	1.0	5-10 Miles	Pacific	41.0	Yes

```
# Mengubah data biner menjadi angka
df['Gender'] = df['Gender'].map({
    'Male': 1, 'Female': 0
})

df['Marital Status'] = df['Marital Status'].map({
    'Married': 1, 'Single': 0
})

df['Home Owner'] = df['Home Owner'].map({
```

```
    'Yes': 1, 'No': 0
})

df['Purchased Bike'] = df['Purchased Bike'].map({
    'Yes': 1, 'No': 0
})

# Mengubah kategori menjadi kolom angka
kolom_kategori_banyak = [
    'Education',
    'Occupation',
    'Commute Distance',
    'Region'
]

df = pd.get_dummies(
    df,
    columns=kolom_kategori_banyak,
    drop_first=True,
    dtype=int
)

print("Data setelah Encoding:")
display(df.head())

print("\nTipe data:")
print(df.dtypes)
```

Data setelah Encoding:

	Marital Status	Gender	Income	Children	Home Owner	Cars	Age	Purchased Bike	Education_Graduate Degree	Education_High School	...	Occupation_Manag
0	1	0	40000.0	1.0	1	0.0	42.0	0	0	0	...	
1	1	1	30000.0	3.0	1	1.0	43.0	0	0	0	...	
2	1	1	80000.0	5.0	0	2.0	60.0	0	0	0	...	
3	0	1	70000.0	0.0	1	1.0	41.0	1	0	0	...	
4	0	1	30000.0	0.0	0	0.0	36.0	1	0	0	...	

5 rows × 22 columns

```
Tipe data:
Marital Status                    int64
Gender                            int64
Income                          float64
Children                        float64
Home Owner                        int64
Cars                            float64
Age                             float64
Purchased Bike                    int64
Education_Graduate Degree         int64
Education_High School             int64
Education_Partial College         int64
Education_Partial High School     int64
Occupation_Management             int64
Occupation_Manual                 int64
Occupation_Professional           int64
Occupation_Skilled Manual         int64
Commute Distance_1-2 Miles        int64
Commute Distance_10+ Miles        int64
Commute Distance_2-5 Miles        int64
Commute Distance_5-10 Miles       int64
Region_North America              int64
Region_Pacific                    int64
dtype: object
```

```
from sklearn.model_selection import train_test_split

# Memisahkan fitur dan target
X = df.drop('Purchased Bike', axis=1)
y = df['Purchased Bike']

# Membagi data 80% training dan 20% testing
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
# Menampilkan jumlah data
print("Jumlah Data Training:", X_train.shape[0])
print("Jumlah Data Testing:", X_test.shape[0])

print("\nPembagian data berhasil!")
```

```
Jumlah Data Training: 800
Jumlah Data Testing: 200

Pembagian data berhasil!
```

```
# Import library
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Membuat model Decision Tree
model = DecisionTreeClassifier(random_state=42)

# Melatih algoritma menggunakan data training
model.fit(X_train, y_train)

# Melakukan prediksi menggunakan data testing
y_pred = model.predict(X_test)

# Menghitung tingkat akurasi
akurasi = accuracy_score(y_test, y_pred)

print("Hasil Evaluasi Algoritma Decision Tree")
print("--------------------------------------")
print(f"Tingkat Akurasi Algoritma: {akurasi * 100:.2f}%")
```

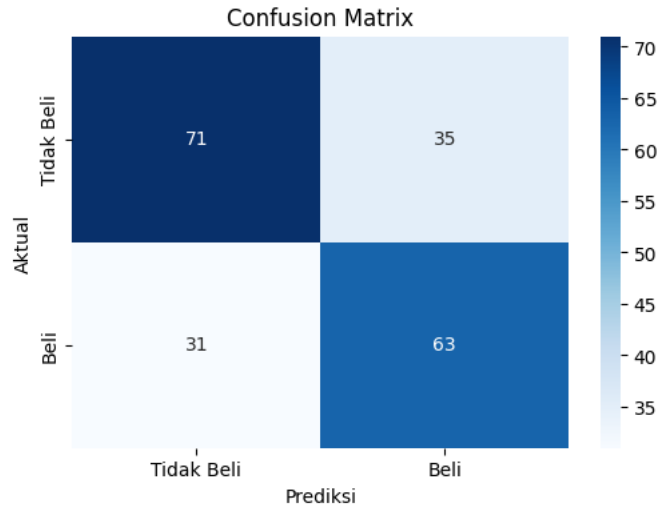
```
Hasil Evaluasi Algoritma Decision Tree
--------------------------------------
Tingkat Akurasi Algoritma: 67.00%
```

```
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['Tidak Beli', 'Beli'],
    yticklabels=['Tidak Beli', 'Beli']
)

plt.xlabel('Prediksi')
plt.ylabel('Aktual')
plt.title('Confusion Matrix')
plt.show()
```

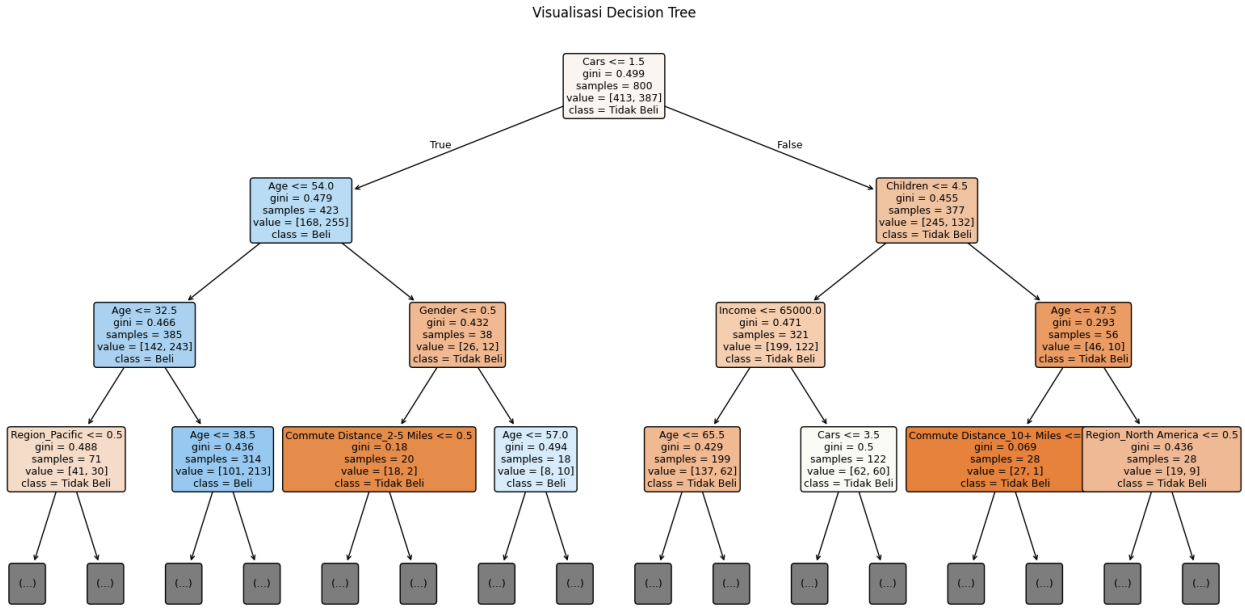



```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(20, 10))
```

```
plot_tree(
    model,
    feature_names=X.columns,
    class_names=['Tidak Beli', 'Beli'],
    filled=True,
    rounded=True,
    max_depth=3,
    fontsize=9
)

plt.title('Visualisasi Decision Tree')
plt.show()
```




```
df.to_csv('bike_buyers_bersih.csv', index=False)

from google.colab import files
files.download('bike_buyers_bersih.csv')
```